

Trevor Sakwa Timothy

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Personal Profile

Mechanical engineer with strong technical foundation in design, simulation, manufacturing optimization, and project management. Supported with a dual master's in Mechanical Engineering with management, and Finance and Investment Management offering a unique blend to evaluate projects through both technical and commercial aspects. Experienced in designing and maintaining systems for reliability, regulatory compliance, and delivering operational improvements. Seeking a Mechanical engineering role where technical expertise and practical problem solving can contribute to innovative engineering solutions.

Skills

Design and simulation

- **Computer Aided Design software:** Advanced SolidWorks, Advanced PTC CREO, Advanced Autodesk Inventor, and intermediate Siemens NX
- **FEA and simulation:** Advanced ANSYS Fluent, Intermediate OpenFOAM, and intermediate Siemens STAR-CCM
- **Programming:** Advanced python, MATLAB, and C++

Manufacturing and optimisation

- **Hands-on manufacturing:** CNC machining, 3D printing (FDM/SLA), laser processing
- **Quality assurance:** Non-destructive testing (ultrasonic inspection, eddy current testing, and visual inspection)

Project and data management

- **Advanced Excel:** pivot tables, dynamic dashboards, and financial modelling
- **Advanced Power BI:** interactive dashboards, data visualisation and commercial performance reporting.
- **Advanced SQL:** querying, data extraction and structuring large data sets for analysis
- **Reporting:** SSRS reporting, Microsoft Fabric, Microsoft project and DAX measures.

Stakeholder co-ordination and communication

- **Cross-functional working:** experience working across operational, financial, and commercial teams to deliver analytical outputs.
- **Clear communication:** translating complex quantitative findings into useful reports and presentations for non-technical members.

Education

MSc Finance and Investment Management | University of Liverpool - Sept 2025- Sept 2026 (present)

- **Key modules:** financial reporting and analysis, Quantitative methods and economics, financial regulation and ethics, corporate finance.
- **Research project:** "Does Cross-market trading volume asymmetry between derivatives and spot markets increase vulnerability to price manipulation? Jane Street case study in India"

MSc Mechanical Engineering with Management (Eng) | University of Liverpool - Sept 2024- Sept 2025

- **Key modules:** Advanced Fluid Mechanics and Aerodynamics, Advanced Laser Materials Processing, Structural integrity, Project management, Computer Aided Design, Nuclear technologies.
- **Research project:** "3D printing of high-performance BASF polymer materials: process optimisation for stronger parts"

Aim: Investigating the effects of design parameters on mechanical properties of 3D printed materials to optimise performance for engineering applications.

Results: Achieved 15% improvement in tensile strength and load bearing capabilities whilst lowering material costs through systematic process calibration.

BEng Mechanical Engineering | Swansea University - Sept 2021- June 2024

- **Key modules:** Finite Element Method, Fluid mechanics, Kinematics and Programming for Robotics, Circuit Analysis, Control systems, Heat transfer.
- **Research project:** “Renewable Energy Solutions for Residential Applications”

Aim: Designing integrated solar, thermal and wind powered system for off-grid power generation.

Experience

Maintenance engineer | Crown Beverages Limited | June 2023- Aug 2023

- **Process optimisation:** Developed Python-based inventory tracking system improving parts and raw material management efficiency by 20% contributing to increase in production output.
- **Structural and Mechanical Integrity:** conducted non-destructive evaluation on production pipework using ultrasonic inspection, and conducted repairs resolving recurring line delays and reducing downtime by 50% translating to direct commercial performance gains.
- **Machinery maintenance:** maintained industrial rotary screw compressed air systems for carbonated beverage production lines, including troubleshooting via HMI diagnostics.
- **Cross-functional coordination:** collaborated with production, logistics, and management teams to align maintenance scheduling with commercial output targets and resource management. Partnered with electrical engineering teams to implement power redundancy solutions.

Engineering Technician | Mount Elgon Hospital | June 2022- Sept 2022

- **Equipment maintenance and reliability:** performed preventative maintenance and calibration on diagnostic equipment namely Philips Intelli Space MRI.
- **Failure Analysis:** conducted fatigue life testing on hospital beds using Eddy current testing thus providing data-driven replacement recommendations.
- **HVAC system design:** Assisted in designing a dedicated HVAC system for newly constructed extension to surgical ward, contributing through simulations and calculations for airflow, filtration, and pressure control in line with ASHRAE 170 requirements.
- **Supplier and stakeholder coordination:** liaised with suppliers and internal departments to ensure timely, accurate information flows across operational and commercial functions. Ensured adherence to SOPs and medical device regulations.

Hands-on Engineering Projects

Swansea University Race Engineering (SURE) – Formula Student Vehicle

- Participated in design and building of a single seat racing car for annual formula student competition where I conducted CAD design of vehicle components using SolidWorks.
- Applied CFD simulation using ANSYS Fluent to perform aerodynamic modelling and optimization to reduce drag.
- Applied FEA to assess structural integrity and fatigue life predictions over the vehicle’s lifespan.
- Coordinated design with manufacturing constrained to ensure all components were optimized for CNC machining and hand assembly through working directly with manufacturing team

Liverpool Waters Waterfront Regeneration project | project management assignment master’s level.

- Performed critical path analysis and detailed work breakdown structures using Microsoft project to identify schedule critical activities, analyze slack and opportunities for schedule compression.
- Developed comprehensive risk registers identifying key project risks while conducting detailed cost analysis to evaluate project financing and cost control mechanisms.
- Managed coordination across construction disciplines, mechanical systems, electrical infrastructure and finishing demonstrating understanding of how engineering decisions cascade across project systems.